



Economics of Electrowinning Iron from Ore for Green Steel Production

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Abstract

The transition to green steel production is pivotal for reducing global carbon emissions. This study presents a comprehensive techno-economic analysis of various green steel production methods, including hydrogen reduction and three different electrolysis techniques: aqueous hydroxide electrolysis (AHE), molten salt electrolysis, and molten oxide electrolysis (MOE). By comparing process flow diagrams, capital and operational expenditures, specific energy consumption, and production footprint, this work provides a high-level assessment of the economic viability of these processes as they mature. The analysis reveals that MOE, despite its ongoing development, offers a promising route for iron production given its ability to process a wide range of ore qualities and the potential to sell electrolyte as a cement product. However, the best balance between deployment ready technology and economic benefit is AHE. Operational challenges are also discussed, such as electrolyte loss and slag handling. We suggest that the sale of by-products like oxygen may not significantly impact the economics due to market saturation. The findings underscore the importance of continued research and development in process optimization to realize the full potential of green steel technologies. All the calculations have been released as supplementary electronic material (MS Excel workbook). The format has been inspired by the techno-economic assessment template (TECHTEST) distributed by the US Dept. of Energy.

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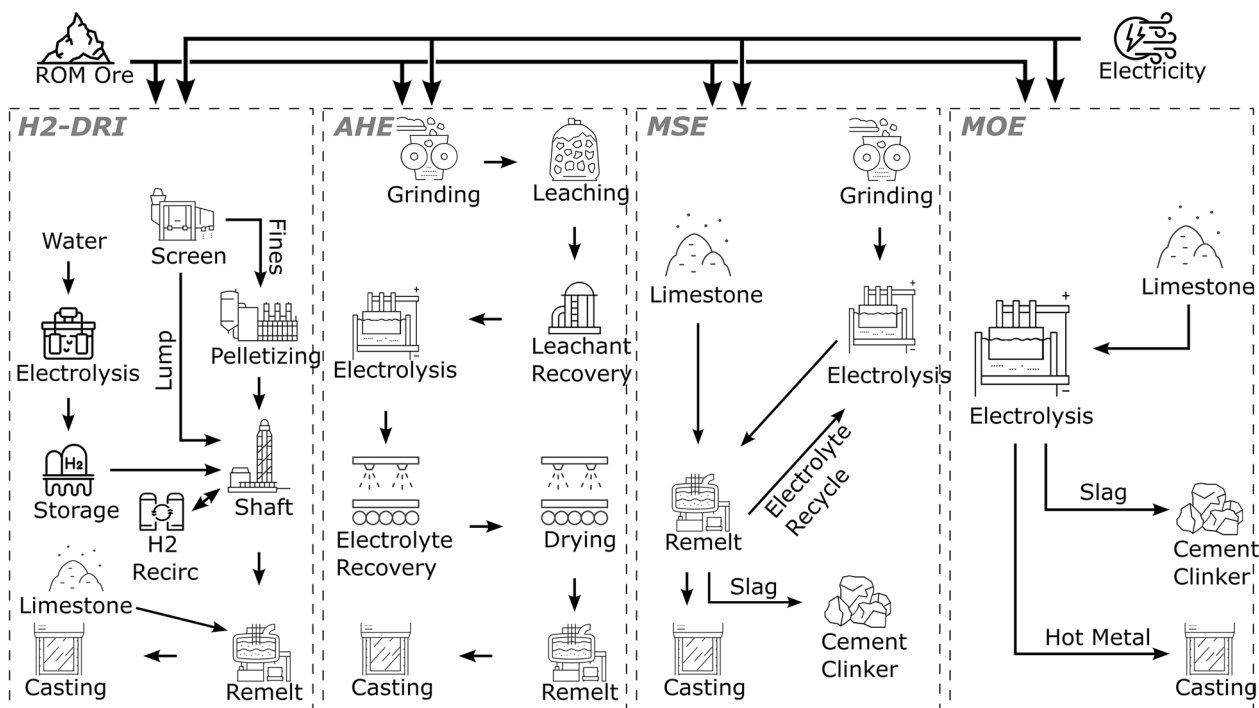
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Graphical Abstract



Keywords Techno-economic analysis · Green steel · Iron electrolysis

Abbreviations

BF	Blast furnace
CAPEX	Capital expenditure
EAF	Electric arc furnace
H ₂ -DRI	Hydrogen direct reduced iron
MOE	Molten oxide electrolysis
OPEX	Operational expenditure
ROM	Run-of-Mine
TPA	Metric tonnes per annum

Introduction

Iron ore reduction in blast furnaces accounts for approximately 7–9% of stationary anthropomorphic carbon dioxide emissions. Steel (an alloy of iron and carbon) is a vital component of human infrastructure, with more than 1.8 billion metric tonnes of virgin material being generated a year. Thus, there is a strong need to continue producing steel and to decarbonize the process. There are several ongoing green steel efforts to mitigate CO₂ emissions from the ironmaking step, where ore is converted to metal. All these possible green iron/steel production methods require additional fundamental research to become viable. These

challenges are outlined in the following section. Additional information on the impact and necessity of decarbonization can be found elsewhere [1–6].

Continued research requires careful consideration of the potential outcomes. To ensure continuing interest in research it is important to evaluate the economics of green steel production as a mature technology. Currently, there are two proposed routes for green field zero carbon iron production: hydrogen reduction and electrolysis. Electrolysis can be further subdivided based on electrolyte; aqueous hydroxide electrolysis (AHE), molten salt electrolysis (MSE), or molten oxide electrolysis (MOE). This paper will present process flow sheets for these electrolysis approaches and compare them with a hydrogen direct reduction (H₂-DRI). This plant scale view provides a comparison of the economics of these processes if they can overcome their current technological and logistical challenges. The aim of this techno-economic analysis (TEA) is to help decision-makers envision these processes as mature and then develop plans to connect current efforts to future results.

Critical to this discussion will be the comparison of four major metrics: capital expenditure (CAPEX), operational expenditure (OPEX), specific energy consumption

Table 1 Challenges for green steel production routes

Technology	Specific challenges	Global challenges
H ₂ -DRI	Hydrogen loss Hydrogen storage Long term maintenance Exhaust gas handling from wet alkaline dust Hydrogen embrittlement of piping Slag handling in subsequent steelmaking	Phosphorous in metal
AHE	Electrolyte (sodium) loss Abrasion from slurry pumping	Supply of green electricity Decreasing ore quality
MSE	Electrolyte (salt) recovery Slag handling	
MOE	Anode materials Tapping metal and electrolyte	

(a component of OPEX), and production footprint. Capital expenditure and operational expenditure will be reported in US dollars for the year 2022. Specific energy will be reported as $\text{kW}\cdot\text{hr kg}^{-1}$ as is standard in the aluminum industry [7]. Production footprint is defined as the ratio of a plant's total annual production, in tonnes (1000 kg) per annum (TPA), to the total plant footprint. Production footprint will have units of TPA m^{-2} of metal. The following sections will compare the process flow diagrams for the three electrolysis processes and the H₂-DRI process and present estimates for CAPEX, OPEX, specific energy, and production footprint.

Overview of Current Challenges of Low Carbon Steelmaking Technologies

Current iron and steel production relies on both the thermal and chemical energy released by the oxidation of carbon. To remove carbon from the process, and keep coal in the ground, new processes must find alternative methods for reducing iron oxide and melting iron metal. Fortunately, there are promising alternative electrical reduction and melting technologies. Yet, there are still unanswered questions regarding how to convert Run-Of-Mine (ROM) ore to metal. These challenges are presented in Table 1.

Of note in the list of challenges presented in Table 1 is that no technology has successfully addressed the issue of phosphorus oxide present in the ore. This is of particular interest to Australian mining operations that are expected to produce ore with increasing phosphorus content (> 0.15 wt%) [8]. A review of H₂-DRI melting and dephosphorization has been recently published that concludes that lower metallization results in lower phosphorus in the produced metal [9].

Some authors have noted that oxygen from electrolytic iron reduction or water electrolysis may be sold as a secondary product [10]. This has not been considered in this analysis because it is believed that the volume of oxygen generated at the scale of a steel plant will drive the price down considerably, reducing the price of oxygen to the cost of transportation. The 2019 US production of oxygen gas was 10.335MT [11]. If annual US steel production (80.5MT) was processed by electrolysis 34.82MT of oxygen would be produced as a co-product. This assumes that the ore is primarily Fe₂O₃. It is expected that the transition to green steel making will take decades, so firms who enter the market early may be able to on-sell oxygen. As the industry transitions to green steel and more oxygen enters the market, the price is expected to fall considerably.

The CIRCORED plant in Trinidad seemingly overcame the specific challenges for H₂-DRI noted in Table 1 [12–14]. However, these sources note damage to refractory linings, possibly because of exhaust humidity. The CIRCORED plant also omitted hydrogen storage, using entirely brown hydrogen for their process. The short operation of the plant did not allow enough time for a full evaluation of hydrogen embrittlement to equipment or other potential long-term maintenance issues. Additionally, the CIRCORED plant produced briquetted iron with a metallization of $\sim 95\%$, leaving downstream remelting furnaces to deal with greater slag volumes than those found in scrap remelting.

The electrolysis processes are at lower Technology Readiness Levels (TRLs) and have not addressed specific operational challenges. This work attempts to quantify electrolyte loss for the AHE and MSE processes, but this analysis is speculative. Additional maintenance and operational challenges should be expected for these processes, and these will need to be explored in future work.

Process Flow Sheets and Boundaries

Figure 1 depicts Process Flow Sheets (PFS) of the H₂-DRI, AHE, MSE, and MOE processes. This type of figure is alternatively referred to as a Process Flow Diagram (PFD). A notable omission from the AHE process is the use of water as an input for leaching and electrolyte recovery. This omission is for clarity. The price and availability of freshwater varies by region, for example many mining operations in the Pilbara region of Western Australia struggle to source enough fresh water to operate. Electrowinning tanks suffer losses from bubble-generated mist and evaporation. Likewise, ore washing is generally not performed in water scarce regions. As these technologies develop, water use should be more thoroughly considered. Conditions such as ore quality, local experience, and electricity generation profile will impact both CAPEX and OPEX. Region-specific costs or opportunities will need to be added to this analysis to determine the appropriate technology for a greenfield ironworks. Additional reviews of these processes, which do not consider balance of plant, are presented elsewhere [3, 4, 6].

The process flow diagrams presented here terminate with hot metal prior to ladle metallurgy. This is likely where phosphorous removal will take place. Remelting is needed for AHE, MSE, and H₂-DRI, but *not* for MOE, as the metal is produced in liquid form. Casting and rolling processes are omitted because they are incumbent processes and would not be affected by changes in upstream technology. While

downstream of the hot metal stage is agnostic to the reduction route, i.e., H₂-DRI or electrolysis, the upstream technology may require modification. In blast furnace ironmaking, the ore may be pelletized or sintered depending on grade. Lower grades generally require additional pre-processing before reduction. Auxiliary equipment both up and downstream from reduction adds costs that are considered here. Quantification of these costs for process comparison has not previously been discussed in literature.

The process flow sheet for H₂-DRI is based on the previous work [10]. H₂-DRI processes currently operating or under construction, the HYBRIT and MIDREX processes, both make use of shaft-type furnaces and have similar flow sheets to that presented by Vogl et al. and displayed in Fig. 1 as the H₂-DRI process. The CIRCORED process operated in Trinidad from 1999 to 2004 and again in 2006 made use of two fluidized beds and produced 300,000 tonnes of HBI during operation [13]. These processes have been investigated and presented elsewhere [3, 6, 15]. In Korea, HyRex (POSCO) and HyCube (Hyundai Steel) processes also use fluidized bed reactors to produce DRI [16, 17]. In no previous process flow sheets there has been mention of grinding ROM ore prior to hydrogen reduction. The CIRCORED literature notes that lump is directly fed into the shaft furnace, hence an additional grinding step is omitted from the PFD presented here.

Molten Oxide Electrolysis (MOE) has been under investigation for more than 100 years and is currently being

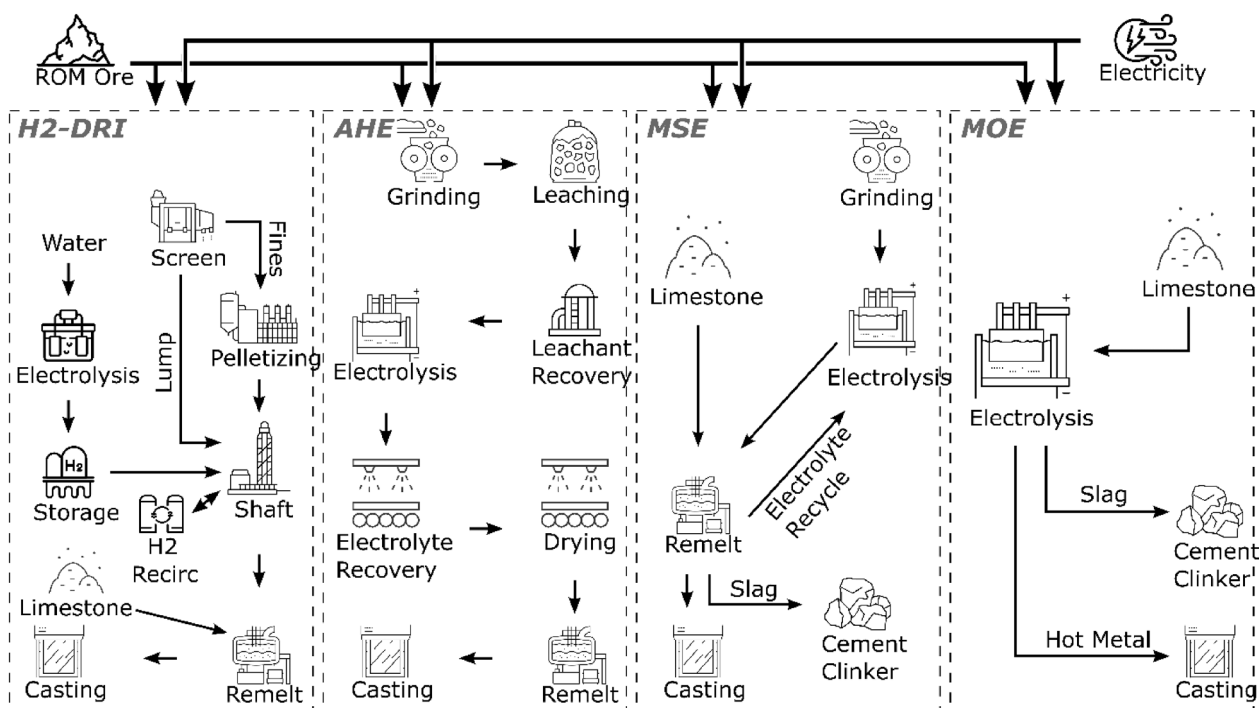


Fig. 1 Process flow diagrams for green steel technologies

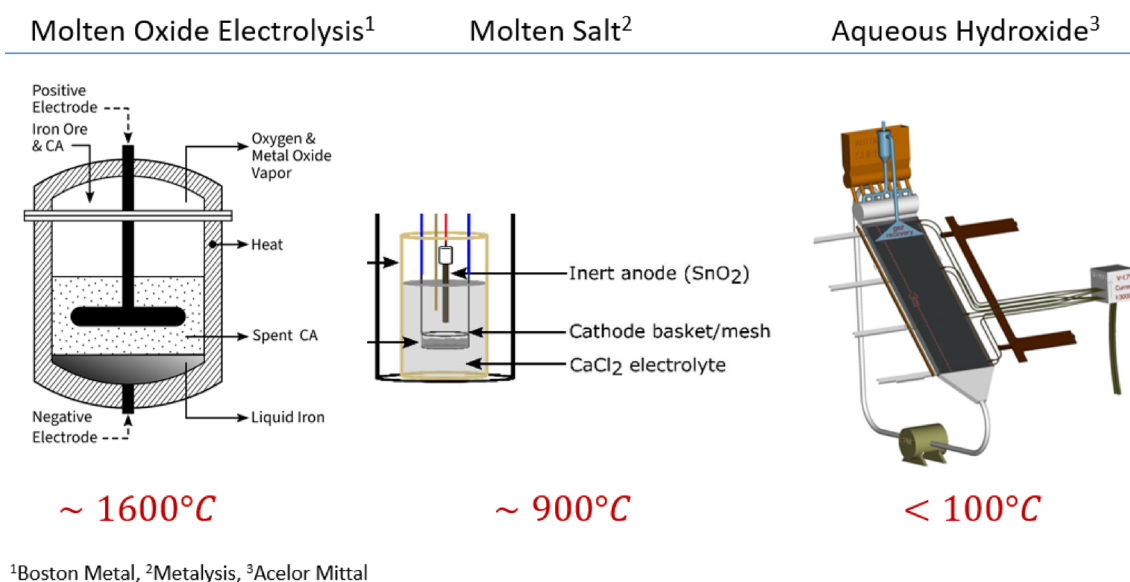


Fig. 2 Illustrations of electrolysis reactors that produce iron from ore. (1) Author's own work derived from Boston Metal website; (2) From Lomax et al. [22] and (3) From Hervé [28])

commercialized by Boston Metal using an electrode material developed at MIT in 2013 [18, 19]. The PFC for MOE comes from images published on the Boston Metal website [20]. Molten salt electrolysis for iron production has been under serious investigation since 2000, with work originating from Cambridge University [21]. We note that the MSE process was originally developed to produce titanium metal from titanium oxide. The process is not commonly considered for terrestrial iron or other, “low value” metals production, but is a leading candidate for lunar in-situ resource utilization (ISRU). Molten salt electrolysis is included in this section because it represents a class of electrolyte that is anhydrous and operates at a temperature between the boiling point of water and the melting point of iron. Specifically, molten salt electrolysis encompasses molten halides [22, 23], carbonates [24], hydroxides [21], as well as ionic liquids [25, 26]. Electrolysis in this middle temperature range reduces the decomposition voltage and some electrolytes may have an increased solubility of iron oxide when compared to aqueous solutions, though this has not been demonstrated. The electrolysis of iron III oxide in aqueous sodium or potassium hydroxide has been investigated by ArcelorMittal starting around 2008 [27]. Subsequently, AHE has become a joint EU project by combining the Ultra-Low CO₂ Steelmaking and ULCOWIN projects to become Siderwin [28]. Figure 2 notes the relative operating temperature of each of the electrolysis processes examined and their associated primary research or commercial projects. Other reviews for green steel production can be found elsewhere [3, 4, 6, 29]

General reactor schematics are illustrated in Fig. 2. The reactor schematic for MS electrolysis omits refractory. There

has been little discussion of the heat balance for these systems. For this work, MSE will use values based on aluminum electrolysis, which operates at a similar temperature (~ 900 °C) with a similar electrolyte. MSE, with a lower operating temperature, can be operated in a steel vessel with insulation or active cooling as needed. The specific design of MSE reactors is beyond the scope of this work but follows similar principles to the design of aluminum electrolysis cells. Sodium hydroxide plants typically operate near 100 °C. Conversely, MOE requires finely balancing the interelectrode spacing or Anode–Cathode Distance (ACD); the resistive properties of the electrolyte; and the current density to control the internal temperature of the reactor (~ 1600 °C).

H₂-DRI

There are several notable components in the H₂-DRI diagram that are not fully evaluated by Vogl et al. Hydrogen storage is noted as a piece of equipment that will allow buffering of load, but its CAPEX is not included in the overall plant model. The capital cost of twelve hours of hydrogen use stored in tanks is included in this work. Additionally, gas treatment is not evaluated thoroughly. The off gas from a hydrogen shaft furnace will be a steam and H₂ mixture. Vogl et al. note that energy can be captured in a condenser/preheater but fail to discuss wet alkali dust handling. Moist dust will form a cake on filters and can be very difficult for traditional baghouses to manage. A 2017 analysis of the MIDREX shaft process by Sarkar et al. provides useful information on the gas composition of a shaft-type furnace

[30]. They present exhaust gas values with concentrations of ~ 25% water vapor. The balance of the gas being ~ 14% CO, ~ 17% CO₂, ~ 33% H₂, with trace amounts of methane and nitrogen. They calculate that the carbon emissions of a natural gas-fired MIDREX plant are 1.269 kg_{CO₂} kg_{Fe}⁻¹, while for a traditional blast furnace with basic oxygen furnace converter this value is 1.87 kg_{CO₂} kg_{Fe}⁻¹. The CIRCORED plant using methane and brown hydrogen produced ~ 0.9–1.0 tCO₂/t_{steel} [13]. While current MIDREX process installations do have higher moisture in the off gas than a blast furnace, they are not operating with pure hydrogen reduction gas. Operating with pure hydrogen will result in the off gas being a mixture of H₂ and water. It is unclear what the hydrogen concentration will be. A simple heat exchanger condenser has been included in the CAPEX estimate used here. A pressure swing adsorption module has also been sized in the TEA. These modules are only 0.5 and 1% of the price of the shaft furnace, and thus the uncertainty in their sizing is within the approximation ranges used in this work. This is a challenge for all hydrogen-based iron reduction processes and raises the question of how completely the auxiliary equipment required for a hydrogen ironmaking process has been considered.

The PFS Vogl et al. present also omits ore processing prior to the shaft furnace. Their schematic indicates that iron pellets are to be fed to the furnace. As noted previously, H₂-DRI tests have been conducted with pellets, sinters, finely crushed material in fluidized beds, and other iron mineral morphologies. Information published on the operations of Outokumpu Technology's Trinidad plant indicates that there were several pre-processing options tested for agglomerating iron ore prior to feeding into the reduction reactor [13, 14]. Both pelletization and micro-granulation are mentioned, but it is unclear which is preferred. The economic analysis presented by Vogel et al. omits the carbon emissions, as well as the capital and operational costs of a pelletizing plant. Auxiliary equipment needs are an open question for both academic and commercial investigations.

The depiction of the H₂-DRI process in Fig. 1 omits the heat exchanger/condenser put forward by Vogl et al. That work does not quantify the performance of this heat exchanger. It is worth noting that most alkali electrolyzers operate at near 100 °C, so it is unclear how much energy can be transferred from the shaft furnace off gas to the incoming hydrogen.

The schematic in Fig. 1 also omits slag from the remelting step in the H₂-DRI and MSE processes. Vogl et al. do indicate that lime addition should be included at a rate of 50 kg per metric tonne of steel produced. This addition rate is similar to that used in a BF. Without flux additions, the slag in the remelt furnace will be viscous and glassy. Flux additions will increase the slag volume and require additional energy to melt. Electric slag furnaces (ESF), an alternative

remelting reactor, have been in operation for several years. A prime example is New Zealand's Glenbrook Steelworks currently, operated by Bluescope [31]. The sands processed at Glenbrook are high in vanadium and vanadium compounds are a high-value product for the plant. Researchers in New Zealand are working toward understanding how to augment the Glenbrook process with MOE to upvalue the slag from the ESF [32, 33].

AHE

The process flow sheet for Aqueous Hydroxide Electrolysis is constructed by inspection of publications from the Siderwin project [28, 34, 35]. Project literature does discuss leaching, but only as a vague requirement. The AusIMM *Cost Estimation Handbook* contains a paper on leach system design with suggested methods for estimating capital and operational expenditures. CAPEX and OPEX estimates following the AusIMM approach agree well with estimates found published by the Siderwin project.

Leaching is primarily performed on ores that have low concentrations of high value metals such as copper and gold. These ores tend to have PPM concentrations of the target metals. For iron ores, the goal would be to leach out impurities of silica and alumina while leaving behind the iron oxide. Generally, iron and its complexes are more soluble than silica and alumina, so the design of the leaching process will require a substantial amount of additional work.

The AHE electrolyte is a high molality sodium hydroxide solution. If sodium hydroxide is used as a leachant both sodium hydroxide and aluminum hydroxide will form. This is, in essence, the Bayer process. These species will need to be precipitated prior to being fed into the reactor because these species will increase the viscosity of the electrolyte reducing efficiency by increasing voltage. This is discussed in more detail in section “Beneficiation”

The PFS also notes electrolyte recovery and drying. These steps require much lighter equipment and much lower energy than any other step in the graphic. While they are critical, it is believed, based on information from copper and zinc electrowinning, that the process of washing and drying cathode plates is largely automated. This will certainly need to be the case for iron production with its higher production volume.

The electrolysis hardware developed during the Siderwin project is unique. The counter flowing gas electrolyte cell is purpose built and highly sophisticated. This is substantially different than the current design of copper and zinc electrowinning plants. Operating electrowinning plants are constructed from polymer concrete with electrodes suspended from bus work. This allows easy electrode changing and inexpensive construction. It is yet unclear how the Siderwin cell will be automated and how fabrication costs will be managed.

MSE

The MSE PFS is largely speculative. Metalysis and the researchers who have developed the process have not published material discussing the balance of plant equipment. Given the high boiling point of the calcium chloride electrolyte, here it is proposed that the easiest method of recovering the electrolyte is to tap it from the remelting furnace. The additional energy to melt the salt from the cathode sponge is difficult to characterize. The specific energy to heat calcium chloride from room temperature to 1900 K is $0.4395 \text{ kW}\cdot\text{hr kg}^{-1}$. The specific energy to heat and melt iron to 1900 K is $0.67 \text{ kW}\cdot\text{hr kg}^{-1}$. If the cathode plates contain only one percent calcium chloride by weight the additional cost of remelting will be minimal. As the concentration of chlorides increases costs may become prohibitive.

It has been shown that alumina and silica are able to be reduced in the chloride-based electrolyte [22]. During MSE, reduction oxygen is removed at the anode while chlorine serves to shuttle the calcium cations back from the anode to the cathode. This effectively lowers the oxygen partial pressure at the cathode causing oxide species to reduce in order of their affinity for oxygen from lowest affinity to highest, this is also referred to as following the electrochemical series where more noble elements are preferentially reduced over less noble elements. As such, in industrial operations where iron ore is being reduced, nearly all the iron will be reduced prior to the reduction of silica to silicon. This transition will happen when the activity of iron oxide is equal to the activity of silica at the cathode.

Demonstration of iron production using molten salt electrolysis was carried out on lunar regolith simulant, a basalt-like silicate mineral with an iron concentration of $\sim 5 \text{ wt\%}$ [22]. The metallic product of that experiment was not remelted, but 96% recovery of total oxygen from the oxide charge was reported. This high oxygen recovery indicates that most of the oxides were converted to metals, the largest metallic fraction being silicon. For terrestrial ores with $\sim 6\%$ silica and alumina, where remelting occurs in air, the silicon and aluminum in the metal product will likely revert to oxides. Additionally, fine oxide particles remaining in the calcium chloride during remelt may limit the ability of the electrolyte to be recycled.

The PFS also indicates that the slag processed during remelting, after the addition of limestone, can become cement clinker. Chloride ions have an accelerating effect on cement and reduce the final strength. As a result, it is unclear if the remelt slag will be of sufficient quality to be used as a cement clinker additive.

MOE

Omitted from the literature, but presented here, is the idea that MOE electrolyte could be derived from ore gangue augmented by limestone addition, and sold on as a cement product. This idea addresses two challenges; how to economically handle gangue, and what to do with it after processing. While this is a novel idea, experimental work needs to be conducted to develop an anode that is appropriate for the system. As such, the sale of cement as a co-product has been eliminated from this assessment.

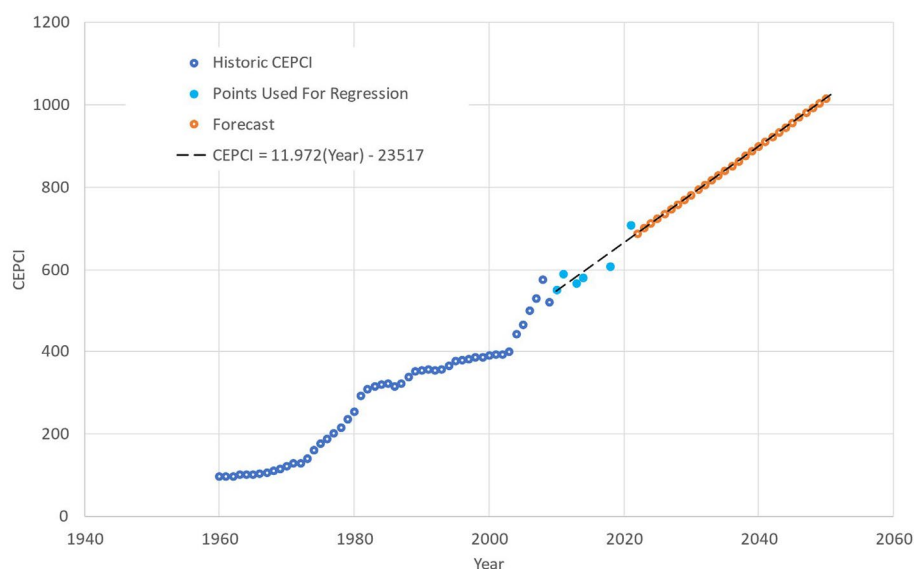
Coupled with the ability to naturally process high gangue ores, MOE also benefits from being able to accept both fines and lumps. The electrolyte used in aluminum electrolysis, cryolite, has a limited solubility of aluminum oxide. This and a density difference between the oxide and metal of only 0.2 g cm^{-3} contribute to problems of “mudding,” where aluminum oxide forms a sedimentary layer on the surface of the aluminum metal. This disrupts cell operation and usually requires fluxing that subsequently volatilizes out of the bath. The large density difference between the electrolyte and metal in an MOE reactor, on the order of 3 g cm^{-3} , and the near infinite solubility of other oxides mean that lumps will be nearly neutrally buoyant and dissolve quickly. Unlike other high-power melting processes, MOE has a relatively low power density of 50 kW m^{-2} . This is calculated by multiplying the electrode voltage of 5 V with an ideal current density of 10 kA m^{-2} . The lowest power density submerged arc and electric slag furnaces operate at approximately 100 kW m^{-2} . Conventional EAFs operate with a hearth power density of $2000\text{--}2500 \text{ kW m}^{-2}$ [36]. A lower power density correlates with less violent bath stirring and gas evolution. Lower gas evolution means that fine ore particles will be able to settle on the bath and dissolve.

The ability to feed fine ore particles, below $45 \mu\text{m}$, is an advantage over other processes. The MOE process flow sheet is simpler which has the potential to reduce both CAPEX and OPEX. Additionally, with one process step there are no bottlenecks from stockpile to hot metal. The process is relatively modular, and with limestone flux addition, it should tolerate a wide range of ores.

Capital Expenditure

Capital cost estimation for these processes is hampered by the opacity of the final design of the electrolyzers. Additionally, capital costs are by nature proprietary. In any case, efforts have been made to source data from several publications to justify assumptions. This section compares the capital equipment costs for the full conversion of ore-to-iron and estimates the electrolyzers using several assumptions. A first estimation for AHE and MSE electrolyzer CAPEX is made

Fig. 3 CEPCI historical data with regression for future forecasting



by treating these components as furnaces operating at the AHE and MSE temperatures. An additional approach uses data for non-ferrous electrowinning, omitting aluminum production, to benchmark electrolyzer pricing. Bulk handling of material, comminution, electrolyte separation, and remelting all have associated CAPEX costs and are accounted for individually.

As a project proceeds from concept toward completion, more granular CAPEX estimates must be made. The primary standard for chemical plant cost estimation is the Association for the Advancement of Cost Engineering (AACE) Recommended Practice No. 17R-97, titled “Cost Estimate Classification System.” This standard provides guidelines for applying the general principles of estimate classification to project cost estimates, which are used to evaluate, approve, and/or fund projects. Estimated CAPEX values used here are sourced from the following references:

- Turton et al. *Analysis, synthesis, and design of chemical processes*. Pearson Education, 2008.
- Woods, D. R. *Rules of Thumb in Engineering Practice*. John Wiley & Sons, 2007.
- Towler, G. and Sinnott, R. *Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design*. Butterworth-Heinemann, 2021.
- *Cost Estimation Handbook*. 2nd ed (2012). Carlton: AusIMM Carlton.

All CAPEX estimates need to be normalized by a price index. The index used in the above references is the Chemical Engineering Plant Cost Index (CEPCI). The CEPCI value is used to convert the purchasing power of dollars in one year to another by the following formula:

$$C_2 = C_1 (I_2/I_1). \quad (1)$$

Here, C_1 is the purchased cost, and I is the cost index with the subscript representing the original year (1) and target year (2). CEPCI data have been extracted from published data [37, 38]. To forecast future CEPCI, a simple linear model was fit to the data between 2010 and 2021. This is displayed in Fig. 3. This model is incorporated into the Techno-Economic Analysis released with this work in the electronic supplementary material as an MS Excel Workbook.

It is crucial to note that there is no mention of electrolysis in the above references except for the AusIMM *Cost Estimation Handbook*. While electrolyzers for metal production are less common than chemical plants, the chloro-alkali process is ubiquitous across the world, and thus its omission is surprising. None the less, these references do capture many of the auxiliary processing operations and provide a consistent framework to evaluate green ore-to-iron processing.

Process Flowsheet Costing

From Fig. 1 diagram of process flow sheets, an inventory of equipment can be tabulated for each process. To ensure even comparison with incumbent technology, all processes are compared at a production rate of 2MTPA. This is of the same order of magnitude as the currently operating integrated steel mills in Australia. The Port Kembla steelworks, operated by Blue Scope, Ltd., produces 2.6MTPA (with a full capacity of 5MTPA) on a 7.6 km² site in New South Wales. The Whyalla steelworks, operated by Infrabuild Ltd., produces 1.2MTPA on a 10 km² site in South Australia. The global average plant size is 3.6MTPA based on data published data [39].

For this analysis, electrolyzers are treated as furnaces with a specified power requirement. In the references above, total power is used to characterize the heated reaction vessels. This approximation is appropriate because, once the ore is inside the electrolyzer, it is processed solely by the passage of electric current. AHE experiments have used nickel anodes and iron cathodes. The nickel anode is very thin and is not expected to significantly contribute to the CAPEX approximation [34]. MSE experiments have used stainless-steel cathodes and tin oxide anodes [22]. Tin oxide, while being inert in chloride melts, is expensive and is likely not durable enough for continuous operations. Affordable and durable electrode materials are the key area of active development for this technology. An estimate of anode cost using tin oxide has been made for the MSE process in the TEA calculations included as supplementary material. By extrapolating from the anode requirements of a copper electrolysis plant, the anode costs for an MSE plant using tin oxide anodes come to an additional \$1.59 per metric tonne of ironmaking capacity. This estimate is for 1.25 m² anodes 15 mm thick with a tin oxide price estimated at \$2000 per metric tonne. The number of anodes for this plant design is high because the current densities of the MSE process are approximately 10 times less than for MOE. The proposed anode material for MOE is chromium based. By scaling the anode area of an aluminum smelter to an MOE plant, both of which are expected to have the same current density, an estimate can be made for the weight of total chromium needed. This is also included in the TEA and adds a total of \$8.37 per metric tonne of capacity to the CAPEX. The cathode for the MOE cells is a bath of liquid iron contained in an electrically conducting hearth which is connected to the negative terminal of the rectifier. This is a standard configuration for DC EAFs and is included in the estimate for electrolyzer cost.

Because the bonds between oxygen and iron are stronger at lower temperatures, there is a higher voltage associated with breaking these bonds and separating the oxygen and iron. The valence of iron, how many bonding electrons it has donated, also impacts the power required for reduction. Iron requires an addition of two or three electrons to become metallic (neutral). FeO requires two electrons while Fe₂O₃ requires three electrons. Fe₃O₄ is a blend of these other oxides and requires an intermediate number of electrons. In this analysis, H₂-DRI and MSE are processing Fe₂O₃, and AHE and MOE are processing FeO. The leaching in AHE and the temperature of MOE pre-reduce Fe₂O₃ to FeO.

The electric current for the plant is set by the production rate requirement. It should be noted that the electrical current needed to produce 2MTPA in an MOE cell is two-thirds of the value for MSE processing. This is a result of the destabilization of the Fe³⁺ ion at temperatures above ~1200 °C in a silica-bearing molten oxide mixtures. Inside

the MOE reactor only Fe²⁺ is stable and this reduces the current requirement.

The voltage requirement is more complicated. For the AHE and MSE processes where the electrolyte is relatively inviscid and thermal balance is more easily achieved, due to lower operating temperatures, the cell voltage, the voltage between the electrodes, is approximately the theoretical decomposition voltage (at that temperature and pH). An operating potential of 1.7 V has been reported for AHE processing from the Siderwin project. This is likely because in the Siderwin reactor water is being split, and the hydrogen is reducing iron oxide to iron metal. The MSE process is generally operated at a fixed potential just above the decomposition voltage of calcium chloride, 3 V. The changing surface area of the MSE cathode can cause high current densities, and thus these reactors are prone to anode effect. Anode effect in molten chloride systems releases chlorine gas damaging equipment and is a current path that does not lead to metallic reduction. For the case of MOE where the electrolyte is more than three times as viscous, and as a result, the resistance of the cell higher, the relationship between operating voltage and decomposition voltage is less clear. In an MOE cell, the primary current distribution dominates, and thermal energy generated between the electrodes overshadows the energy consumed by electrolysis. The voltage estimate used for AHE is 1.7 V, the decomposition voltage of water is at a pH of 15 (1.2 V), plus an additional 0.5 V of electrode overvoltage. This matches the reported voltage by the Siderwin project. For MSE the voltage is set at 3 V, as reported. For MOE the decomposition voltage of 1.64 V is used with an overvoltage of 2.58 V based on estimates from Judge and Azimi [40]. With the current and voltage set their product, the reactor power, is used to estimate furnace capacity.

Another key component of many proposed green steel processes is the remelting of the iron after reduction. This is cited in the work by Vogl et al. for H₂-DRI and Draxler et al. for other green steel processes. The primary reference for estimating the CAPEX of the electric arc furnace is a Boston Consulting Group report [41]. This source puts forward a CAPEX value of €184/t crude steel (211.03USD(2022)/t) for an electric arc furnace. The derivation of this value is not explained. Given that many green(er) steel routes rely on remelting accurate CAPEX and OPEX of electric arc furnaces should be developed further. The value presented in the BCG report is in line with an estimate that can be derived from a 2020 commissioning of a US Steel EAF and tubular mini-mill in Fairfield, Alabama. This plant is reported to have required a \$412 M (2020USD) investment and will produce 1.6MTPA of tubular products [42]. By implementing Eq. (1) using data from Fig. 3 for 2010 and 2021, the predicted CAPEX for the Fairfield EAF is \$256.74 (2021USD)/t CS, or just over half of the plant cost. The news

article discussing this plant noted that some of the equipment was recommissioned from other plants. Thus, the value presented in the BCG report is appropriate for this analysis.

The references used to examine the balance of plant costs in this section do not contain information on electrolyzers. Data for hydrogen production can be used to estimate the cost of the Siderwin style AHE electrolyzer. Mergel et al. cite values of $\sim \text{€}1000 \text{ kW}^{-1}$ (in 2013) of installed alkaline electrolyzers [43]. This is the source of data used by Vogl et al. This electrolyzer cost can be multiplied by the theoretical power to arrive at a value of $\$291.74 / \text{t CS}$ (2018 dollars) for an ideal system and $\$331.82 / \text{t CS}$ (2018 dollars) for the reported specific energy consumption of $2.72 \text{ kW}\cdot\text{hr kg}^{-1}$. To estimate the electrolyzer CAPEX for MSE and MOE processes, data from the aluminum smelting industry can be used. In aluminum, smelting high-performance refractories and a graphite bottom are used in the construction of each cell. These materials drive up the construction cost. It is presumed that the higher costs for aluminum cell construction are at parity with the construction costs for MSE or MOE reactors. The work by Stinn and Allanore discussed in the following section places the CAPEX for an aluminum electrolyzer between $\$5000$ and $\$10,000 \text{ USD}(2018)/\text{tonne}$ [38]. This is slightly higher than the value of $\$3,205 \text{ USD}(2018)/\text{tonne}$ cited by Tabereaux [7]. When these CAPEX are scaled by the room temperature metal density, the estimated CAPEX for MSE and MOE electrolyzers fall between $\$1715 \text{ USD}(2018)/\text{t CS}$ and $\$3430 \text{ USD}(2018)/\text{t CS}$.

Extrapolation from Other Electrolysis Processes

A recent review of capital expenditure costs for electrowinning processes has been presented by Stinn and Allanore [38] based on the AACE method presented by Towler and Sinnott [44]. They examine copper, magnesium, aluminum, and chlorine production. They derive a common capital expenditure formula using process specifics, such as product density, by fitting a power law function to real plant cost data.

The application of Stinn and Allanore's approach to the AH, MS, and MOE processes is reported in Table 2. These values are reported as dollars of installed capacity per metric tonne of annual production. The regression performed

by Stinn and Allanore did not consider process steps such as comminution or leaching. Additionally, because of their choice of regression, the high processing temperatures cause the α_1 & α_2 pre-factors for front-end and electrolysis processing to grow very large for both the MSE and MOE; potentially overestimating the CAPEX for these processes.

Comparison of Different Estimates

The CAPEX estimates for green steel technologies presented by Draxler et al. omit discussion of the upstream and downstream processing requirements presented [6]. The CAPEX estimates by Draxler et al. are presented as 2022 US dollars per metric tonne crude steel in Table 3. The values derived from the regression formula presented by Stinn and Allanore are in reasonable agreement with AHE processing and dramatically overestimate the processing cost for MSE and MOE processing. This is believed to be a result of the large temperature-dependent pre-factor used in the “pre-processing” term. That term likely captures aluminum anode production well but provides limited extensibility for green steel plants. The AHE estimate arrived at in this work is in good agreement with the value presented by Draxler. The largest uncertainty in this analysis is the cost of the electrolyzer. For AHE, this is estimated as an open stainless-steel tank and amounts to $\sim \$131/\text{tonne}$. For MOE, the electrolyzer price is calculated by scaling CAPEX pricing for aluminum electrolysis projects from 2020 by the room temperature density difference. It is unclear how much the electrolysis unit drives the cost of the aluminum plant. MSE electrolyzer pricing comes from the Capital Cost Database and is taken as a molten salt fired heater. Given that fired heaters are lower in cost than electric heaters this value is likely underestimated.

CAPEX is largely driven by the price of the electrolyzer. Given the low TRL of the electrolysis process, the values reported here are likely underestimated. For example, piping, pumps, filters, and associated balance of plant are not captured for AHE and MSE. The CAPEX estimate for an MOE plant internalizes much of the balance of plant because of the process similarity with aluminum electrolysis. The second largest capital expenditure is the EAF remelting furnace. The shaft furnace for the H_2 -DRI process is also a large component of CAPEX at $\$240/\text{tonne}$. This shaft furnace price comes from a recent process sensitivity analysis

Table 2 Summary of capital equipment costs using the method developed by Stinn and Allanore [38]

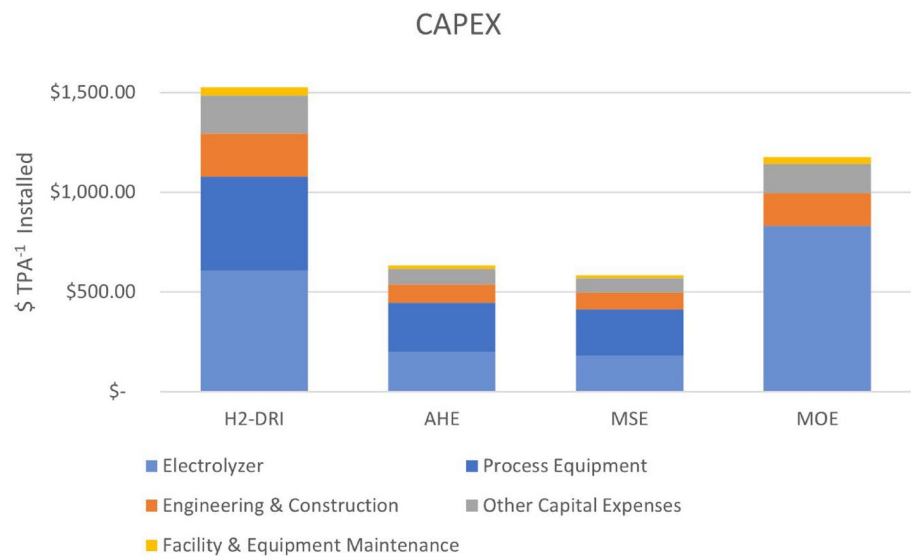
Component	AHE	MSE	MOE
Front-End	\$325.26	\$2,063.87	\$2,734.57
Electrolysis	\$227.79	\$7,650.47	\$1,113.76
Rectifier	\$523.68	\$1,232.50	\$1,053.53
Total	\$1,076.74	\$10,946.85	\$4,901.86

Values are reported in $\$/\text{TPA}$

Table 3 CAPEX estimate comparison

Process	Draxler	Stinn & Allanore	Shahabuddin et al	This work
H_2 -DRI	\$175.84		\$750.00	\$1442.55
AHE	\$611.61	\$1076.74	\$890.00	\$632.31
MSE		\$10,946.85		\$583.90
MOE	\$764.52	\$4901.86	\$1,125.00	\$1,163.81

Fig. 4 Capital expenditure for green steel processes in this work



performed by Bhaskar et al. [1]. This value is in line with full plant costs for MIDREX plants in Toledo, Ohio and Corpus Christi, Texas reported as \$453/t and \$414/t in 2022 dollars, respectively. From the *Cost Estimation Handbook*, the shaft furnace is estimated to be 50% of the total plant cost. Engineering and construction costs, other capital expenditures, and facility and equipment maintenance costs are all estimated as a percentage of initial purchase price. This breakdown is shown in Fig. 4.

Operational Expenditure

The three main components of OPEX are labor, electricity, and raw materials. The price of ore, a raw material, for each process is the same. The additional energy for upgrading or modifying the ore is included as “plant” electricity to separate it from the electricity used for electrolysis. The electricity used to remelt the iron in an electric arc furnace after reduction is also calculated separately.

Plant Labor Productivity

Towler and Sinnott present estimates for the number of shift positions per process step in Sect. 8.5.1 of their work [44]. They estimate that 4.8 people should staff each shift position to cover three shifts per day, along with weekends, holidays, personal time off, and sick leave. This product is shown in Eq. (1). The method presented by Towler and Sinnott is also used in the TEA template provided by the US Department of Energy, TECHTEST [45]. Figure 5 is reprinted from [44]. Using this diagram, for each process step, we can estimate the number of shift positions then multiply by 4.8, the

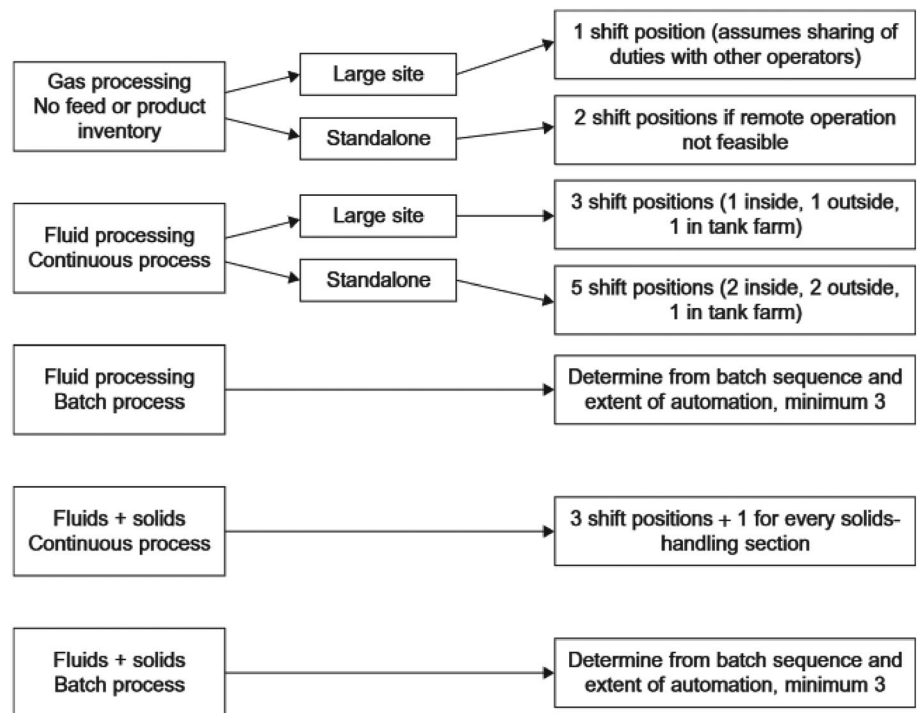
number workers per shift position, and then multiply the total by 2000 h per year. We repeat this process for each process step in the process flow diagram to find the total number of person-hours per year. We can then divide the total person-hours by total production in tonnes to arrive at the person-hours needed to produce a tonne of steel for each process. The average OECD labor rate is then used to determine the \$/tonne of labor that is included in OPEX.

$$\sum_{\text{Process Steps}} \text{Shift Positions} \times 4.8 \left(\frac{\text{People}}{\text{Position}} \right) \times 2000 \left(\frac{\text{hrs}}{\text{yr}} \right) = \text{Total Person} \cdot \text{hrs} \quad (2)$$

To estimate the labor costs for each processing route, the individual components are considered. The estimated shift positions are displayed in Table 4. Again, the primary uncertainty lies with the electrolyzer. Data from zinc and aluminum plant staffing have been used to estimate the number of people required for each potline. The number of potlines is estimated following guidance from the *Cost Estimation Handbook* [46]. Estimates for the H₂-DRI process are in line with the staffing of MIDREX plants. MOE staffing is estimated from staffing of Australian aluminum electrolysis plants. In an aluminum electrolysis plant roughly 1/3rd of the people are employed on the potline with approximately ¼ being employed in the anode plant. The remainder of staff are assigned roughly equally to maintenance, services, casting, admin, and contractors. There are slightly fewer admin and slightly more contractors than this even split [47].

The electrolyzer staffing estimates in Table 4 are likely underestimated. Plant automation will be critical to reducing

Fig. 5 Number of shift positions by process step. Reprinted from [44]



labor costs. The electrolyzer in the H_2 -DRI process produces hydrogen and oxygen. This is a common industrial process that involves only fluids and gases. It is largely automated as well as continuous but does require reasonable maintenance. The EAF used in the MSE and H_2 -DRI processes includes one additional shift position over the EAF staffing estimate from the AHE process. This additional position is to handle charging from bulk storage and to help manage the higher slag volumes from this process. The AHE EAF will be able to charge cathode plates, and this is anticipated to be largely automated.

In an aluminum plant each potline may have 4–8 operators per shift, with many plants having four to six potlines. The Al Taweelah smelter has six potlines. Aluminum electrolysis requires frequent anode changes and tapping each with a team of operators. Similarly, copper electrolysis, conceptually similar to AHE, requires frequent anode and cathode changes, but because of the lower weight of the copper anodes and cathodes these teams can be smaller and use lighter cranes.

Labor in OECD countries can be very high. The International Labour Organization has tabulated labor costs by country and normalized by purchasing power parity (PPP) and 2017 exchange rates. In 2017, the labor rate (\$/hr) for Australia was \$31.69 and Norway was \$61.51. Average OECD countries fall in between these values. The labor cost for each process is estimated using the Australian labor rate and the shift position values from Table 4 and recorded in Table 5 as \$/t CS in 2022. The 2017 labor rate has been

Table 4 Shift positions for green steel plants

	Equipment	Type	Positions	Total per shift
AHE	Ball mill	Continuous	4	77
	Slurry mixer	Batch	3	
	Cone roof tank	Batch		
	Cone roof tank	Batch	3	
	Precipitation	Batch		
	Electrolyzer 5/potline	Batch	60	
	Blower	Continuous	4	
	Conveyer	Continuous		
MSE	EAF	Batch	3	52
	Ball mill	Continuous	4	
	Cathode prep	Continuous	4	
	Electrolyzer 5/potline	Batch	40	
MOE	EAF	Batch	4	24
	Electrolyzer 6/potline	Semi-Batch	24	
H_2 -DRI	Electrolyzer	Continuous	5	15
	Tanks		2	
	Condenser			
	Shaft furnace	Continuous	4	
	HBI storage	Batch	4	
	EAF	Batch		

Table 5 Labor cost per metric tonne of hot metal produced reported as USD(2022)/t HM

	Estimate	Source	Estimate	Source
H ₂ -DRI	\$11.13	MIDREX staffing	\$8.43	Towler & Sinott (Table 4)
AHE	\$25.83	Zinc smelter	\$43.30	
MSE	\$25.83	staffing	\$29.24	
MOE	\$20.95	Aluminum smelter staffing	\$13.50	

increased by the ratio of Consumer Price Index for 2017 and 2022.

Due to the additional process steps and the associated equipment, the AHE process has higher labor rates than the others. The magnitude of these values depends linearly on the number of shift positions. The shift position estimates from above should be considered a low value given they are derived from chemical plant values and these plants are largely automated while metallurgical plants tend to have less automation. The conclusion from this analysis is that MOE requires much less labor than other electrolysis processes.

Beneficiation

Current commercial ore generally contains approximately six percent by weight silica, alumina, and other gangue species. In a blast furnace, these species are fluxed into a liquid slag by the addition of limestone. For some green steel processes, gangue represents a challenge. Silica and alumina will dissolve in sodium hydroxide solutions increasing the viscosity and retarding iron deposition. In molten salt and hydrogen processes, silica and alumina will prevent reductants, (hydrogen and calcium, respectively) from reaching iron oxide. If gangue is not removed prior to reduction both hydrogen and molten salt processes will need to handle substantial slag volumes during the consolidation and remelting of the iron product.

While gangue is a major challenge for H₂-DRI, AHE, and MSE, it presents a lesser challenge for MOE (in particular in relation to major gangues, e.g., SiO₂ and Al₂O₃) and may be a benefit. MOE can operate using an electrolyte that has a similar composition to blast furnace slag. Thus, the traditional practice of adding limestone to ore as it enters the blast furnace can be carried over to charging MOE cells. Early efforts to understand the electrical behavior of minerals have shown that iron can be reduced from basalt-like minerals [48]. It is yet to be seen how phosphorous behaves in an electrolytic cell with a molten oxide electrolyte, as well as with other green steel alternatives. Here, we presume that the challenges with sulfur present in blast furnace iron production come entirely from the coal and coke used. Manganese, being slightly more nobler than iron may become an

impurity, but this is also the case with the incumbent blast furnace process.

Concurrent with efforts to develop zero carbon steel, world ore quality is declining. All leading steel producers are now dealing with increased gangue. Nunna et al. discuss this trend and present a detailed discussion of current ore grades in their 2022 paper [8]. Their work primarily addresses Australian ores and discusses efforts to keep grade constant with minimal beneficiation. They note that Brazilian ore remains of high quality and that native Chinese ore is of substantially lower quality with nearly 50wt% gangue. The canonical example presented is that of the Christmas Creek and Cloudbreak mines operated by Fortescue Metals Group, LTD. These sites make use of a beneficiation circuit that maintains ore quality and consistency.

Table 6 presents mineralogical process requirements for the blast furnace and two DRI processes [8]. These processes are believed to be a good proxy for H₂-DRI, as the main difference is the substitution of hydrogen for natural gas. The authors note that this is not a one for one change and that there are differences, but the Midrex and HyL processes are well established and provide a reasonable proxy for ore grade requirements particularly when considering the plant built in Trinidad [13, 14]. There are ongoing tests to determine if a fluidized bed reactor, used in the Trinidad plant, or a shaft furnace, as used by the Midrex process, is the better ore reduction approach. A fluidized bed requires a finer average particle size and more narrow particle size distribution, 1–2 mm [13]. A shaft reactor can accept a larger particle, as seen in Table 6, but has a lower cut-off to prevent losses to the exhaust gas.

As mentioned, electrolysis processes have not quantified the challenges of phosphorus, sulfur, or bound water (represented by the Loss-On-Ignition column) impurities. Question marks have been used to indicate this uncertainty. MOE has notably larger tolerances on the quality of ore that can be processed. This is because with the addition of an appropriate amount of limestone, the gangue material becomes electrolyte. Melting the additional gangue material will incur an additional cost, but this may be offset by the sale of cement clinker. The MOE process benefits by having a continually moving reaction surface, the interface between the electrolyte and metal. This allows convective and diffusive transport in the liquids. The AHE and MSE processes

Table 6 Size and impurity requirements for different green steel technologies

Furnace	Physical size (μm)	Specification (wt%)				
		Fe (min)	Gangue ($\text{SiO}_2 + \text{Al}_2\text{O}_3$) (max)	P (max)	S (max)	LOI (max)
Blast Furnace ¹	6000–30,000, max 5% < 6300	62–65	6	0.05	0.01	2.0
DRI						
Midrex ¹	6000–35,000, max 5% < 6300	67	3.2	0.015	0.015	1.5
HyL ¹	6000–25,000, max 2% < 6300	65.5	2.2	0.02	–	1.5
Electrolysis						
AH ²	10	69	< 0.1%	?	?	?
MS ³	53–1000	65	5	?	?	?
MOE ⁴	30,000	30	70	?	?	?

¹From [8]²From [28]³From [22]⁴From [49]**Table 7** Particle size distribution of iron ores reprinted from [50]

Iron Ore	Size (mm)					
	+5	3.5–5	2–3.5	1–2	0.5–1	-0.5
A	32.6	25.1	16.1	14.0	5.0	7.2
B	50.5	13.8	7.7	7.2	9.2	11.6
C	57.9	20.6	10.7	6.4	2.5	1.9
Blend	21.9	18.7	18.7	14.4	7.0	19.4

rely solely on diffusion through the solid material, which is several orders of magnitude slower than liquid diffusion. The result of this is that MOE can operate at higher current densities than AHE and MSE.

Grinding and Leaching Energy

The AHE and MSE processes require the iron ore feed to be ~ 10 μm and 53–1000 μm , respectively. Xiao et al. have published ROM particle size distributions of samples taken from the Wuhan Iron and Steel Co., Ltd. [50]. These are shown in Table 7. With > 20% of the ore being sized + 5 mm is clear that ROM iron ore will require additional grinding for the AH and MS electrolysis processes. MOE will not require further grinding because the ore is completely dissolved in the molten oxide mixture.

The energy to grind the ore from the sizes shown in Table 7 to ~ 10 μm for AHE and 53–1000 μm for MS is 0.064 $\text{kW}\cdot\text{hr kg}^{-1}$ and 0.045 $\text{kW}\cdot\text{hr kg}^{-1}$, respectively. These energy values are based on average equipment. Grinding media selection and equipment specifications will impact

these values. These values were derived from information found in Lichter and Davey [51] and from work on assessing the energy efficiency of the German cement industry [52]. These estimates also agree with values presented elsewhere [53].

Grinding energy presented here also agrees with Bond Work estimates taken from processing tests on two iron ore samples [54, 55]. A complete review of this method is presented in Wills' text [56]. The Bond work index reported for iron ore varies between 10 and 24 $\text{kW}\cdot\text{hr ton}^{-1}$. This corresponds to a total work to reduce the 80% passing particle size from 20000 μm to 10 μm of between 0.031 and 0.074 $\text{kW}\cdot\text{hr kg}^{-1}$.

The values presented above are in line with estimates presented by Barberousse et al. reviewing the potential impact of the Siderwin project for Europe [35]. They estimate that grinding cost will be 0.066 $\text{kW}\cdot\text{hr kg}^{-1}$. Barberousse et al. also report the specific energy from leaching for the AH process will be 0.325 $\text{kW}\cdot\text{hr kg}^{-1}$. They do not cite the source of this value.

Electrolyte Recovery

Previous work has noted that some electrolyte is lost during MSE [57]. Electrolyte loss in the MSE process comes from electrolyte adhered to the surface of the product, but also electrolyte that is present inside the produced “sponge.” Because the MSE operating temperature is lower than the melting point of the metal, voids are left as oxygen is separated. These voids are then filled with electrolyte. The electrolyte can either be washed from the sample or separated during remelting. There are no published estimates on the quantity of electrolyte remaining in the product after reduction.

There are two potential paths for recovering the molten salt electrolyte: melting the sponge and electrolyte or washing the electrolyte from the sponge. Both require estimating the amount of electrolyte that needs to be recovered. Presuming the metal sponge is 65% metal by volume, the rest being calcium chloride, and the metal having a density of $7.87 \text{ g}\cdot\text{cm}^{-3}$, one kilogram of metal sponge will have a volume of 195.48 cm^3 . This will leave 68.42 cm^3 of calcium chloride, which at standard temperature has a density of $2.15 \text{ g}\cdot\text{cm}^{-3}$. The total mass of calcium chloride per kilogram of iron metal is 147.1 g. To heat and melt this mass to 1600°C requires $0.063 \text{ kW}\cdot\text{hr}$. An alternative to melting the retained salt is to wash and dry the sponge. This would involve converting fresh water to brine, recovering the salt from that water, then drying the sponge to an appropriate level for remelting. There will likely be some residual moisture in the sponge, and this will oxidize the metal, reducing either the quality or quantity of the product. Calculating this is left for future work.

The AHE electrolyte is an aqueous sodium hydroxide solution at a concentration of 10–15 M. The product of the AHE process is a nearly fully dense plate, though no density numbers have been published. The electrolyte loss for this process is expected to be minimal because of the low surface area to volume. Again because of the high volume to surface area, residual moisture is not expected to be an issue.

The prices for lost electrolyte are minimal when compared with the total production costs. For AHE, the cost of lost electrolyte is $\$0.63/\text{metric tonne}$. For MSE processing, the cost is $\$0.29/\text{metric tonne}$. These estimates are for prices of $\$500/\text{tonne}$ NaOH and $\$250/\text{tonne}$ CaCl_2 with yearly losses of 10%.

Thermodynamic Specific Energy Consumption

The specific energy consumption to produce iron metal from ore is reported in $\text{kW}\cdot\text{hr kg}^{-1}$ or $\text{MW}\cdot\text{hr tonne}^{-1}$, these units being synonymous. The energy for the decomposition of

hematite (Fe_2O_3) to iron and oxygen is shown in the following reaction:



The energy of this reaction, when proceeding from left to right at 298.18 K is $743.523 \text{ kJ/mol}_{\text{Fe}_2\text{O}_3}$. To calculate the energy by iron metal mass, we divide this energy by the two moles of product, and again divide by the molecular weight of iron. The result, $1.828 \text{ kW}\cdot\text{hr kg}^{-1}$, is the theoretical minimum decomposition energy to produce one kilogram of iron at 25°C . A similar process can be used to calculate the minimum energy needed to reduce FeO to iron metal.

As temperature increases the energy for decomposition decreases; thermal energy reduces bond energy. The MSE process will need to reduce the Fe^{3+} ion found in hematite. There is a small penalty to generate hydroxide ions in the AHE process, but this is accounted for by calculating the energy for the leaching stage separately. As the temperature rises, the Fe^{3+} ion becomes less stable and will decompose to the Fe^{2+} ion. There is a debate on how to calculate and measure the valence of the iron ion in molten oxide solutions. The consensus seems to be that silica in the oxide melt stabilizes the 2^+ valence, while melts that are more basic stabilize the 3^+ valence. Table 8 presents electrolysis with specific energy values calculated from the NIST JANAF data [58]. We can see that the difference in specific reduction for MOE between Fe_2O_3 and FeO is on the order of 10%. This reduction in decomposition energy is compensated by a similar rise in the specific enthalpy required to heat the reactants to temperature. The right most column shows the total energy required to heat the reactants and then perform the reduction. The calculations show the advantage of MOE. MOE requires 25% less theoretical energy than AHE.

Scaled Specific Energy Consumption

The previous section presented ideal thermodynamic reduction energy. This section presents estimates based on extrapolations from existing processes and are meant to include heat and other system losses of the plant at scale. This method takes inspiration from the method for calculating CAPEX presented in Section “Capital Expenditure.” Specific energy for the AHE process is estimated by scaling values from the chloro-alkali and other aqueous electrowinning processes. The MOE and MSE process estimates come from extrapolations based on the aluminum smelting industry. H_2 -DRI values come from the HYBRIT and MIDREX processes.

The blast furnace process emits between 1.8 and 2.2 kg of CO_2 per kg of steel produced. This estimate includes emissions from pelletizing, sintering, coking, ironmaking (blast furnace), steelmaking (basic oxygen furnace), as well

as reheating and rolling. Of note is that using methane or brown hydrogen produces $\sim 0.9\text{--}1.0 \text{ tCO}_2/\text{t}_{\text{steel}}$ for reduction only [13]. While this is an approximate 50% reduction from specific blast furnace emissions, electrolysis with green electricity has effectively zero carbon emissions.

Much work has gone into quantifying emissions from integrated steelworks. A very detailed accounting has been performed by Zhang et al. for a 0.3MTPA Chinese plant [59]. The steelworks audited by Zhang et al. is smaller than the globally most common size of 2MTPA but does contain all the relevant equipment common in larger plants. Zhang et al. calculate that the plant produces rolled products with a carbon footprint of $2.067 \text{ kg}_2 / \text{kg}_{\text{steel}}$. This includes emissions associated with the rolling plant. The rolling plant is omitted from the present analysis and should be revisited by future work. Current rolling mills use excess blast furnace gas, ladle furnace gas, and coke oven gas as fuel for the billet reheat furnaces. Electrically resistive reheat furnaces are uncommon, presumably because of the high cost of electricity.

To compare green steel production, which will consume electrical energy, to the incumbent blast furnace method, which consumes carbon in the form of coked coal, we can divide the CO_2 intensity of a coal fired power plant by the mass specific emissions of an integrated steelworks. This is shown in Eq. (4). The CO_2 intensity for a coal power plant is taken as $0.820 \text{ kg}_{\text{CO}_2} (\text{kW}\cdot\text{hr})^{-1}$. This value comes from data collected by the Intergovernmental Panel on Climate Change (IPCC) [60].

$$2.067 \text{ kg}_{\text{CO}_2} / \text{kg}_{\text{steel}} \div 0.820 \text{ kg}_{\text{CO}_2} / \text{kW}\cdot\text{hr} = 2.52 \text{ kW}\cdot\text{hr} / \text{kg}_{\text{steel}} \quad (4)$$

Equation (4) gives a firm footing to start to compare green steel production routes to incumbent technologies based on a common specific energy metric. The assumption here is that the conversion efficiency of coal to heat and reductant is similar to the conversion efficiency of electricity to the same. For large industrial processes, hydrocarbons and coal are the status quo. While, as evidenced by this work, there has been considerable effort put into electrifying the steel industry, there are many other industrial processes that use natural gas for heat because of the lower associated purchase and operating costs. Because there is a lack of data on large-scale electric heating, the assumption made here that there are similar conversion and environmental losses between fossil fuels and electricity will require further investigation.

The CO_2 intensity of steel from the Zhang et al. estimate includes rolling, which leads to an overestimation of the specific CO_2 emissions for ironmaking and steelmaking. Equation (4) can be compared to the theoretic minimum energy for the reduction presented in Table 8, represented in the row for MOE reduction of Fe_2O_3 . The value of $2.68 \text{ kW}\cdot\text{hr kg}^{-1}$ from Eq. (4) is slightly less than twice that of the heating of

Table 8 Ideal specific energy for different electrochemical routes

Process	Species	Temp (K)	Reduction energy ($\text{kW}\cdot\text{hr kg}^{-1}_{\text{Fe}}$)	Heating reactants	Total
AHE	Fe_2O_3	383	1.80	0.028	1.83
	FeO		1.22	0.021	1.24
MSE	Fe_2O_3	1173	1.29	0.32	1.61
MOE	Fe_2O_3	1873	0.84	0.57	1.41
	FeO		0.77	0.60	1.37

iron oxide to 1873 K and performing reduction, $1.41 \text{ kW}\cdot\text{hr kg}^{-1}$. The difference represents the process efficiency of a BF/BOF plant, $\sim 50\%$.

Because the AH process operates at near 110°C , heat loss is minimal. Thus, the main energy loss comes from the overvoltage on each electrode. In the production of chlorine in the chloro-alkali process, the overpotential is approximately 0.1 V [61]. At 100°C , the decomposition voltage of Fe_2O_3 is 2.59 V , which equates to $1.8 \text{ kW}\cdot\text{hr kg}^{-1}$. The overpotential adds $0.069 \text{ kW}\cdot\text{hr kg}^{-1}$ to the decomposition energy.

The specific energy for MS and MOE production follows by scaling the specific energy reported by the aluminum industry by decomposition voltage, electrolyte resistivity, temperature, density, and valence. Density is scaled by the room temperature density of the metal. The valence for MOE is taken as 2^+ because it is expected that Fe^{3+} decomposes Fe^{2+} at temperatures above approximately 1000°C as Fe^{2+} is more stable [62]. This assertion is an active area of debate for researchers, but as noted elsewhere, in the presence of silica there is strong evidence that Fe^{2+} is the dominant species.

Four factors are used to scale the specific energy consumption of aluminum electrolysis. The ratio of decomposition voltage accounts for the strength of the oxygen to metal bond. The ratio of electrolyte viscosities accounts for electrode overpotentials as well as the solution resistance of the electrolyte. The ratio of temperatures relates the thermal losses to the environment from the reactor. For a self-heating reactor, the energy needed to maintain constant temperature will come from the electrical energy supplied. The assumption here is that thermal losses are proportional to the temperature difference with the environment and are thus linear. While convection is proportional to the difference of the squares of the temperature, this lumped parameter approach has been used for other scaling estimates [63]. The density ratio scales the denominator of specific energy. The ratio of valence accounts for the number of electrons needed per atom of product.

Electrode overpotentials are not easily calculated a priori. An effort has been made to quantify these overpotentials by others [40]. Judge and Azimi do not include an estimate for anode bubble voltage, which is shown to contribute $\sim 5\%$

Table 9 Scaling factors for estimating specific energy consumption of MS and MOE iron production

Property	Electrolysis process		
	Aluminum	MS	MOE
Decomposition voltage (V) ¹	4.47	1.79	0.82
Electrolyte conductivity ($\Omega^{-1} \text{ cm}^{-1}$) ²	2.58	2.55	0.78 ³
Electrolyte viscosity (cP)	6.7	1.87	20
Temperature (K)	1173	1173	1873
Metal density (g cm^{-3}) ²	2.7	7.87	7.87
Valence	3	3	2
Specific energy ($\text{kW} \cdot \text{hr kg}^{-1}$)	13–20	0.50–0.77	2.89–4.45

¹From[58]²From[67]³From[68]

to the total cell voltage of an aluminum electrolyzer [64]. Vogt presents that electrode overpotentials can arise from both insulation of the electrode by bubbles and mass transfer limitations to the electrode [65]. The movement of species to the electrode and the movement of bubbles away from the electrode are dominated by viscosity through the Eyring relationship for diffusion and the Navier–Stokes equations for fluid flow. Chmielowiec et al. note the challenges of extrapolating the behavior of one electrochemical system to another based on differences in the electrolyte properties, electrode surface roughness, and temperature [66]. As no clear theory exists for predicting system behavior at scale, we have elected to use viscosity as a representative quantity for inter electrode phenomena. Scaling values are presented in Table 9.

The base specific energy for aluminum smelting is between 13 and 20 $\text{kW} \cdot \text{hr kg}^{-1}$. The lower value comes from [69]. The upper value is taken from data published by the Al Taweelah Smelter.¹ This smelter is rated at 3500 MW and produces 1.5MTPA of aluminum metal. The value of 20 $\text{kW} \cdot \text{hr kg}^{-1}$ calculated from the Al Taweelah plant should be considered an upper limit. The onsite power generation also supports a desalination plant, which uses a substantial amount of electricity. The scaling of these values to MS and MOE is presented in the following equations:

$$13 - 20 \frac{\text{kW} \times \text{hr}}{\text{kg}_{\text{Al}}} \times \frac{1.79 \text{ V}}{4.47 \text{ V}} \times \frac{1.87 \text{ cP}}{6.7 \text{ cP}} \times \frac{1173 \text{ K}}{1173 \text{ K}} \times \frac{2.7 \text{ g cm}^{-3}}{7.87 \text{ g cm}^{-3}} \times \frac{3}{3} = 0.50 - 0.77 \frac{\text{kW} \times \text{hr}}{\text{kg}_{\text{Fe}}} \quad (5a)$$

$$13 - 20 \frac{\text{kW} \times \text{hr}}{\text{kg}_{\text{Al}}} \times \frac{0.82 \text{ V}}{4.47 \text{ V}} \times \frac{20 \text{ cP}}{6.7 \text{ cP}} \times \frac{1873 \text{ K}}{1173 \text{ K}} \times \frac{2.7 \text{ g cm}^{-3}}{7.87 \text{ g cm}^{-3}} \times \frac{2}{3} = 2.89 - 4.45 \frac{\text{kW} \times \text{hr}}{\text{kg}_{\text{Fe}}} \quad (5b)$$

The result of Eq. (5a) for MSE being lower than the decomposition energy is a result of the need for additional external heating for this system. The viscosity is linearly related to conductivity, as noted by Zhang et al.[70], and the use of a calcium chloride electrolyte means that additional heat will need to be supplied to keep the system from freezing; there will not be enough Joule heating to keep the system molten. Thus, the minimum additional heat needed to keep the system operating will be equal to the difference between the value calculated in Eq. (5b) and the ideal decomposition voltage presented in Table 9.

The above estimation agrees well with values presented by Judge and Azimi [40]. They estimate a specific energy consumption of 4.78 $\text{kW} \cdot \text{hr kg}^{-1}$. They use an “inside-out” approach by accounting for interelectrode phenomena then adding buswork and plant losses at the end. They validate their approach by performing a similar accounting for an aluminum electrolysis cell and recovering a good match to empirically reported values. As shown in Table 10, electrolysis is the major driver of energy consumption for green steel production. This is analogous to the blast furnace being the main source of carbon in the incumbent process. The iron–oxygen bond requires the most energy to break. Heating, pre-processing, and transportation are small in comparison.

Ore heating has also been added to the MSE process. This additional heat requirement is again a result of the MSE process requiring additional heat input. In aluminum electrolysis and MOE, ore is brought to temperature by the excess Joule heating. The scaling procedure above does not capture this subtlety. Here, the MOE and aluminum electrolysis process are more closely related, and this is how the scaling theory has been derived. This scaling process may require additional refinement for applicability to MSE.

Summary of Specific Electrical Energy Consumption

Specific energy consumption for each process step for the four green steelmaking technologies proposed is presented in Table 10. Though these processes are quite different, a reasonable attempt has been made to compare them. Finer granularity in energy accounting may increase the fidelity of these calculations. The upper value presented here by scaling data from the aluminum industry is in good agreement with an inside-out summation of energy losses presented by Judge and Azimi [40]. They arrive at an estimated specific energy consumption for MOE of 4.7 $\text{kW} \cdot \text{hr kg}^{-1}$.

¹ Google maps link—<https://tinyurl.com/mu7dhjyd>

Table 10 Total specific energy consumption

Process step	H ₂ -DRI ³ kW · hr kg _{Fe} ⁻¹	AHE	MSE	MOE
Grinding ⁴	–	0.07	0.01	–
Leaching ²	–	0.33	–	–
Pelletizing ¹	0.11	–	–	–
H ₂ production (electrolysis)	2.41	–	–	–
Ore heating	0.32	–	0.32 ³	–
Electrolysis (ideal)	–	1.33 ²	1.29 ⁴	0.79 ⁴
Electrolysis (ideal + enthalpy) (Low)	–	1.869 ⁴	1.81	2.89
Electrolysis (ideal + enthalpy) (High)	–	–	2.08 ⁴	4.45 ⁴
Electrolysis (reported)	–	2.72 ²	–	–
Electrolyte recovery (melting)	–	–	0.06 ⁴	–
Remelt (low)	0.75	0.52 ²	0.52 ²	–
Remelt (high)	–	0.67 ³	0.67 ³	–
Total (low)	3.59	2.781	2.720	2.89
Total (high)	–	3.779	3.138	4.45

The AH total value has used the reported value for electrolysis provided by the Siderwin project [28]. The above extrapolation from chloro-alkali electrolysis to the Siderwin system underestimated the magnitude of the electrode overpotentials. This is likely because the electrodes are much closer in the chloro-alkali process than in the Siderwin process. Thus, there may be higher solution resistance, or because oxygen is being evolved on the anode, the gas phase kinetics may be different.

Consumables and Electricity

Electricity dominates the OPEX for green steel production. Vogl et al. use a value of \$33.23/MW·hr in 2022. As shown in Table 10, the specific energy for green steel production is approximately 3 kW·hr kg⁻¹ and this energy must be completely supplied by electricity meaning that the electrical energy cost is ~ \$99.69/t CS in 2022 dollars. The additional costs are consumables for the EAF (including lime for fluxing), electrolyte for the AHE and MSE processes, and the raw ore price. The raw ore price and alloying elements are the same for each process and taken as \$115.92 in 2022 dollars. Vogl et al. estimate lime and EAF electrode costs of 5€/t CS and 8€/t CS both in 2018 values, respectively. The sum of these values is approximately 10% of the electricity cost.

Wear parts and maintenance will be higher for AHE processing. Pumping of abrasive slurry and the corrosivity of

sodium hydroxide will require higher plant maintenance than moving dry bulk material. Towler and Sinnott note that maintenance can usually be taken as 3–5% of installed costs. Presuming the AHE requires more maintenance, and the other processes require less, this is still a small cost when compared to the price of electricity.

For the H₂-DRI process, the cost of hydrogen is the largest factor. Vogl et al. cite Mergel et al. and estimate hydrogen costs between 2 and 4€ kg⁻¹. A 2023 news article reports the US price of green hydrogen as \$5 kg⁻¹ (4.63 € kg⁻¹) [71]. With 53.7 kg of hydrogen needed per metric tonne of steel this quickly becomes the largest cost for this process. This cost comes directly from the energy required to produce hydrogen. With the price of electricity of \$33.23/MW·hr, as cited by Vogl et al., a lower heating value of 33.33 kW·hr kg⁻¹, and an efficiency of 70%, the price of hydrogen is calculated as \$1.58 kg⁻¹. Thus, this analysis is underestimating the current cost of hydrogen by nearly a factor of two. This equates to \$80.12/tonne CS. If the price of hydrogen rises to \$3/kg, the cost of hydrogen per tonne of steel rises to \$160.24/tonne CS. This will drive the overall OPEX of H₂-DRI up to ~ \$334.24/tonne CS. The OPEX estimates for the processes considered are shown in Fig. 6.

Plant Footprint

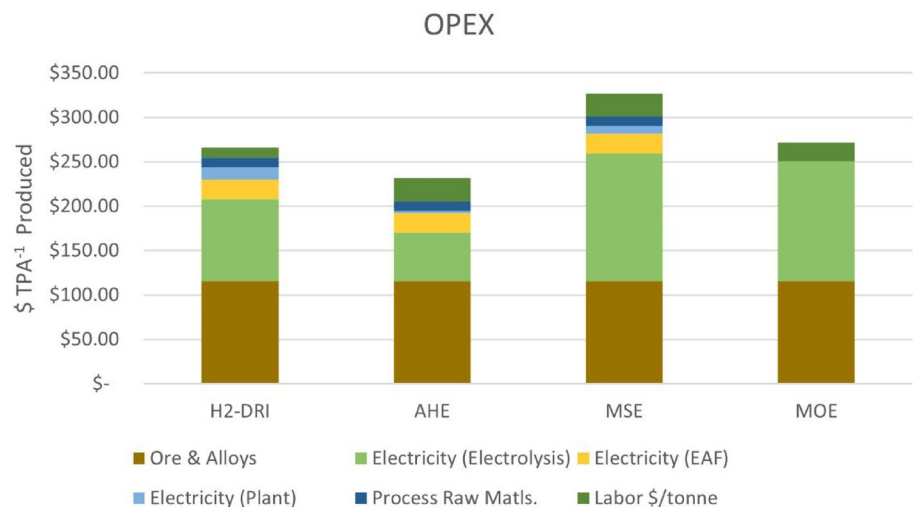
Land area required has been noted as a significant detractor for continued research into electrolysis [72]. To estimate the approximate land size of a 2MTPA green steel plant, site data are taken from a chlorine plant, an aluminum smelter, and the TOHO Zinc electrolytic iron plant. These data are compared with current land use for Australian steel makers.

The AHE process has a similar production footprint to copper electrowinning and iron electrorefining. The electrolytic steel process is analogous to copper electrorefining, where an impure anode is dissolved into the electrolyte to be plated onto a cathode. Little or no gas is evolved in current electrolytic steel or copper production. This contrasts with the AHE process, where primary reduction is carried out in the cell and the anodic reaction releases oxygen.

Hydrogen generation has a similar production footprint with the chloro-alkali process, both using either proton exchange membranes or alkali electrolyzers. The primary difference being that in the chloro-alkali process, chlorine gas and sodium hydroxide liquor are the main products. Water electrolysis generates hydrogen and oxygen, both as gases.

In the AHE process, sodium hydroxide liquor handling and solid iron handling will be different, but the main component of each plant is the electrolysis area. Similarly, the

Fig. 6 Operational expenditure for green steel processing reported in \$/tonne of hot metal. The high electricity cost for MSE results from Fe_2O_3 being electrically reduced, while for AHE and MOE the species is FeO



potlines in an aluminum smelter occupy the largest portion of the site, which is expected in an MOE plant. The extrapolation of aluminum conditions to iron production for the MOE process is more well founded than extrapolating aluminum production to MSE because MOE and aluminum electrolysis are more analogous. The cathode geometry in MSE remains an open design question. The values for MSE presented here are only for completeness and will need further investigation to validate.

The Olin Chloro Alkali Products plant in McIntosh, Alabama was taken as a representative plant for this sizing estimation. The plant is sited on 0.63km^2 and produces 400kTPA Chlorine and sodium hydroxide. An aerial view of the plant is shown in Fig. 7. A 1 km scale bar is shown toward the bottom of the image.

By dividing 400kTPA of sodium hydroxide and chlorine by the sum of their molar masses, we can estimate the molar output of the plant: $5.29 \times 10^9 \text{ mol yr}^{-1}$. Multiplying this molar production rate by the molecular mass of diatomic hydrogen results in a production rate of 10,520TPA of H_2 . Assuming we need 50% additional hydrogen over the stoichiometric value to reduce iron, this plant could supply enough hydrogen to produce 19,583TPA of iron metal. The assumption that additional hydrogen will be needed is also made by Vogl et al. Using the plant footprint of the Olin plant, the size of a hydrogen plant to produce 2MTPA of iron will be 64.34km^2 . This is almost an order of magnitude larger than the other proposed processing plants. Combined hydrogen and smelting plant sizing has not been estimated in the literature.

For AHE processing, where the area of the TOHO Zinc electrolytic iron plant has been used as the scaling factor, the plant size is comparable to existing steel plants. MSE and MOE plants with a production capacity of 2MTPA are expected to be of similar size to the Al Taweelah plant.

Existing plants are described in Table 11. Production data from the Al Taweelah smelter have been scaled by the valence difference between iron and aluminum and density, as has been done.

Sensitivity and Other Costing Aspects

This work should be considered as a conceptual comparison of iron production routes. As such, the accuracy of the values presented here is assumed to be approximately $\pm 50\%$. Yet, the CAPEX of the hydrogen electrolyzer is high. While this technology has the lowest technical risk, it does require large initial capital investment. AHE has the lowest combined



Fig. 7 Olin Chloro Alkali Products McIntosh, AL plant

Table 11 Size and production capacity of integrated steel mills and aluminum smelters

Plant	Scale process	Capacity (MTPA)	Size (Mm ²)	Production density (TPA m ⁻²)	2MTPA sizing (km ²)
Cleveland cliffs	MIDREX	1.6	0.57	2.82	0.71
Port kembla	BF/BOF	2.6	7.6	0.34 ¹	5.85
Whyalla	BF/BOF	1.2	10	0.12	16.67
Gwangyang	BF/BOF	23	19.28	1.19	1.68
Al Taweelah Smelter(Al)	MOE	1.5(Al) [2.9Fe] ²	6	0.25(Al) 0.48(Fe)	4.18
Olin Chloro-Alkali	H ₂ -DRI	0.020	0.63	0.03	64.34
TOHO Zinc	AHE	0.008	0.05	0.15	13.24

¹Port Kembla is currently operating at ½ capacity²Scaled by density and valence

CAPEX and OPEX, for the assumptions made here. The cost of the electrolyzer, which is more sophisticated than current industrial units, may be underestimated. However, even if this price were doubled, the CAPEX for AHE would still be lower than for H₂-DRI. Similarly, the electrolyzer price for MSE is uncertain. Solid product handling, as is done in zinc and copper electrowinning, has not been achieved in baths of ~ 900 °C. This may require sophisticated robotic designs. Conversely, the construction materials for MOE reactors are well understood with many electric slag furnaces and DC electric arc furnaces operating. The main uncertainty of this estimate arises because the CAPEX has been scaled from aluminum electrolyzers and not ferrous metal furnaces. This scaling was chosen because better price information was available for aluminum smelters.

Even though AHE seems to be the best choice as a green steel production route, there are still technical challenges to overcome that could balloon costs for any of these technologies. For H₂-DRI, the largest technical risk is the efficiency and lifetime of the electrolyzer. A good faith estimate has been put forward here but the true cost of current H₂ electrolyzer installations is obscured by government incentives and venture capital funding. For MSE, the uncertainty lies both in the material handling of the cathode, i.e., how are fine particles of ore electrically connected and how are they installed and removed from the cell and the anode material. For AHE, the leaching of millions of tonnes of ore may be problematic. Similarly, the electrolyzer infrastructure may be sophisticated and require large amounts of cathode plate handling. It is uncertain how the thin deposits that have been reported will sum to two million tonnes per annum. Like MSE, the anode material for MOE is also uncertain. Higher value alloys or active cooling of the anode structure may allow the process to operate but will drive up operating costs. With these challenges in mind, the cost of any of these processes may double or triple if no technological solutions

can be found. This is the primary uncertainty for these processes moving forward.

Finally, environmental concerns are not included in this assessment. Public perception and siting of these facilities should be assessed. Hot processes like H₂-DRI and MOE, even with good exhaust treatment, will always have fugitive emissions. AHE and MSE will require storing chemicals that have the potential to leach into groundwater. The municipal regulatory environment may make one or more of these processes difficult to site.

Downstream Challenges

Decarburization and alloying are the penultimate steps in the incumbent steel production process, occurring prior to casting and rolling. Current blast furnace and DRI routes add carbon to the metallic iron during reduction. This lowers the melting point of the metallic iron by nearly 100 °C and increases the fluidity. This is beneficial for remelting DRI in an EAF. Additionally, during decarbonization in a BOF, the charge temperature increases as the oxygen and carbon react. Without the presence of additional carbon EAF, heating times may be longer and additional heat may need to be supplied during alloying and ladle treatment. Other authors have identified and discussed these challenges [2]

Conclusions

This paper concludes that the three electrolysis processes are reasonably comparable, and that hydrogen iron making will require a significantly larger specific footprint, i.e., hectares per tonne of hot metal. The large plant footprint for H₂-DRI is a result of low hydrogen production per unit plant area. With the current cost of hydrogen greater than \$4 kg⁻¹,

hydrogen steelmaking OPEX will be greater than \$400 t⁻¹, the current LME average sale price for iron metal.

H₂-DRI is the most developed technology, as demonstrated by the number of plants under construction. Electrolysis processes lack the technological maturity of H₂-DRI but hold the promise of lower CAPEX and OPEX. This conclusion is contingent on better electrolyzer cost estimation, which is likely proprietary. Unlike the MIDREX process where hydrogen can gradually be substituted for natural gas, there is no intermediate step for electrolysis. However, electrolysis plants are more modular than furnaces, thus, a plant can be designed and commissioned in stages.

Electric slag furnaces provide an operational middle ground between EAFs and MOE reactors. These are currently used to process several sand-based ores using graphite electrodes. These furnaces are used by New Zealand Steel, Highveld Steel, and Steel Dynamics [73]. They can augment the thermal energy from carbon combustion with electrical energy and deliver high carbon hot metal to a decarburizing step. Hopefully, a more complete carbon accounting can be performed on these furnaces to better understand their competitiveness with natural gas fired shaft furnaces that are combined with EAF melting.

This study concludes with the following:

- CAPEX of electrolyzers for all processes is the dominant cost and has high uncertainty
- OPEX, excluding ore, is driven by the cost of electricity and possibly labor
- AHE and MSE have complex flow sheets and will require more labor than H₂-DRI or MOE
- H₂-DRI OPEX is dominated by the price of hydrogen. This work estimates an OPEX price of \$1.5/kg from optimistic literature. A more realistic estimate of \$2–\$3 per kg makes the OPEX of H₂-DRI the highest of the processes considered.
- 2MTPA plant footprints across electrolysis processes are comparable with incumbent technology.

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Declarations

Conflict of interest The authors declare no conflicts of interest with regard to this work.

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